

DNA computing, sticker systems, and universality^{*}

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Abstract. We introduce the *sticker systems*, a computability model, which is an abstraction of the computations using the Watson-Crick complementarity as in Adleman’s DNA computing experiment, [1]. Several types of sticker systems are shown to characterize (modulo a weak coding) the regular languages, hence the power of finite automata. One variant is proven to be equivalent to Turing machines. Another one is found to have a strictly intermediate power.

1. Introduction

The *sticker systems* introduced here are language generating devices based on the *sticker operation*, which, in turn, is a model of the techniques used by L. Adleman in his successful experiment of computing a Hamiltonian path in a graph by using DNA, [1]. We recall some details of the experiment in order to see the roots of our models.

One knows that DNA sequences are in fact double stranded (helicoidal) structures composed of four nucleotides, A (adenine), C (cytosine), G (guanine), and T (thymine), paired A–T, C–G according to the so-called Watson-Crick complementarity. If we have a single stranded sequence of A, C, G, T nucleotides, together with a single stranded sequence composed of the complementary nucleotides, the two sequences will be “glued” together (by hydrogen bonds), forming a double stranded DNA sequence. Figure 1 illustrates this operation.

Using this biochemical reaction, Adleman has proceeded as follows, in searching Hamiltonian paths in a graph:

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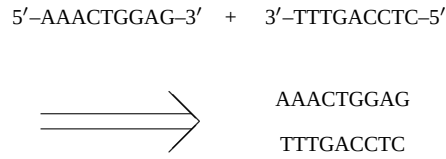


Fig. 1.

- codify the nodes by single stranded DNA sequences of length 20 and put all these strings in a test tube,
- if the node i is codified by the string x_i and the node j is codified by the string x_j , and there is an arrow from node i to node j in the graph, then add to the test tube a single stranded DNA sequence y_{ij} such that if $x_i = x'_i x''_i$, $x_j = x'_j x''_j$, each of x'_i, x''_i, x'_j, x''_j being strings of length 10, then $y_{ij} = y'_{ij} y''_{ij}$, where y'_{ij} is the Watson-Crick complement of x''_i and y''_{ij} is the Watson-Crick complement of x'_j .

Due to the complementarity, the string y_{ij} will match the corresponding parts of x_i and x_j , linking them and producing in this way a sequence of length 40, as illustrated by Fig. 2.

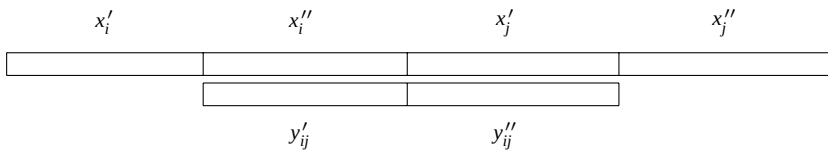


Fig. 2.

This “domino game” can continue, identifying longer and longer paths in the considered graph. By a filtering procedure which is not of interest here, one then can check whether or not paths with specified properties exist (for instance, Hamiltonian paths).

We extract from this experiment only the basic ingredient: the operation of prolonging to the right a sequence of (single or double) symbols by using given single stranded strings, matching them with portions of the current sequence according to a complementarity relation.

The formal model of this operation is the *sticker operation* defined in the following section.

This operation can be used in building a generative/computing device: start from a given set of incomplete double stranded sequences (axioms), plus two sets of single stranded complementary sequences. Iterating the right prolongation using elements of these latter sets, we get “computations” of possibly arbitrary